



2 MILLION

THE ESTIMATED NUMBER OF INDIVIDUAL MOSAIC TILES ORIGINALLY USED TO CREATE THE MADABA MAP

The Madaba map is a mosaic showing Palestine and lower Egypt, with particular focus on towns and other sites of Biblical importance. This part of the map shows Jerusalem.

THE FIRST DESCRIPTION of the **bubonic plague** was given by Roman historian Procopius (c.500–c.565). He was present in Constantinople (now Istanbul) when the disease struck the Byzantine empire in 542. He described the characteristic swellings (or buboes) under the arms and around the groin, and a type of delirium brought on by septicemia (blood poisoning) that caused sufferers to run around screaming.

By the 6th century, the cartographic tradition inspired by Ptolemy was waning, to be replaced by a religiously inspired view of Earth. The **Madaba map**, thought to be the oldest surviving map of Biblical cities, dates back to this time. In around 550, **Cosmas Indicopleustes**, a merchant from Alexandria, composed the *Topographia*



7:10

Bubonic plague death toll

At its peak, the bubonic plague, which struck the Byzantine empire in 542, killed 10,000 people a day in Constantinople alone.

Christiana (*Christian Topography*), which controversially presented the **world as a flat space** dividing the heavens from the underworld, and in which Jerusalem occupied a central position. Cosmas located Paradise just beyond the ocean that surrounded Earth.

One of the leading medical practitioners at the time of the Byzantine emperor Justinian was Alexander of Tralles (c.525–c.605). His *Twelve Books on Medicine* described a range of diseases including those caused by intestinal parasites. He was the first physician to **identify melancholy (depression)** as a cause of suicidal tendencies.

Around 570, Chinese mathematician **Chen Luan** mentioned the abacus for the first time in a commentary on an earlier work of the 2nd century. He described 14 methods of arithmetical calculation, one of which he referred to as “ball arithmetic,” in which a series of wires were suspended on a wooden frame, with four balls strung on the lower half of each wire representing a unit each, and a ball on the upper half representing five units.

Although the Chinese had a **long tradition of canal building**, their

Rainbow bridge

This bridge over a side section of the Grand Canal at Wuxi, China, arches in a dramatic fashion, which gave this type of construction the nickname “rainbow bridge.”



CHINESE BLOCK PRINTING

Printing using wooden blocks was probably invented in China in the 6th century, although the first complete surviving printed book dates to 868. A manuscript was prepared on waxed paper, which was rubbed against a wooden block to transfer a mirror image of the characters onto it. The block was then carved and used for printing.

greatest project was the cutting of the **Grand Canal from Changan to Loyang**, under the Sui dynasty, which joined up earlier, smaller canals. Its main section, the Pien Chu canal, which was 621 miles (1,000 km) long was completed in 605 and was said to have taken five million laborers to build.

By the early 7th century, Chinese engineers had worked out that bridges did not need semicircular arches. In 605,

Chinese engineer Li Chun completed construction of the **Anji Bridge** in Hebei. The arch was flattened by two smaller arches in its spandrels (the triangular area bounded by the outer curve of an arch and adjacent wall), which spread the weight more evenly and meant that only one main arch was needed to span the river.



542 Procopius describes an outbreak of **bubonic plague** in Constantinople

c.550 John Philoponus proposes an early form of **theory of inertia**

542–65 Madaba mosaic map is created in Jordan

c.550 Greek physician Aetius describes **forceps** and breast cancer

c.560 Alexander of Tralles describes **psychiatric conditions**, including depression

c.570 First mention of **abacus**

595 An iron chain **suspension bridge** built in Yunnan, southwest China

c.600 Hohokam people of southwest USA create **large-scale irrigation networks**

605 Construction of the **Grand Canal** in China is completed